

MSBC 5680

Optimization Modeling

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus, Online Section Only)	Data Scientist (77.0 / 100)	71.0 / 100

71

CURRENT AVG SCORE

across 6 job roles

96

POTENTIAL SCORE

with recommended additions

+34

MAX SCORE GAIN

best single-role improvement

CURRENT COURSE DESCRIPTION

Focuses on formulating decision problems as mathematical models and employing computational tools to solve them. Microsoft Excel is used as the main modeling platform but the course will also cover advanced tools, such as modeling languages. Optimization modeling will be illustrated in problems associated with operations, marketing, management, and finance. Integrates topics from decision analysis and operations management as they relate to modeling management decisions.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	77.0	100.0	+23.0	MEDIUM
Applied Scientist	76.9	100.0	+23.1	MEDIUM
Machine Learning Engineer	76.0	100.0	+24.0	MEDIUM
AI Engineer	71.8	100.0	+28.2	MEDIUM
ML Engineer	64.1	81.4	+17.3	MEDIUM
MLOps Engineer	60.4	94.9	+34.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

LLMs & Generative AI 1 of 6 roles

storage

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.
- LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.
- LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Data Scientist

77.0 → 100.0 (+23.0 pts)

data sets insights different social build approaches

Applied Scientist

76.9 → 100.0 (+23.1 pts)

biology research storage technical agents data learning systems

Machine Learning Engineer

76.0 → 100.0 (+24.0 pts)

data sets learning business design development science model

AI Engineer

71.8 → 100.0 (+28.2 pts)

data learning build software technical model systems technology

ML Engineer

64.1 → 81.4 (+17.3 pts)

data learning machine systems model build technical digital

MLOps Engineer

60.4 → 94.9 (+34.5 pts)

data model text systems learning machine training development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.